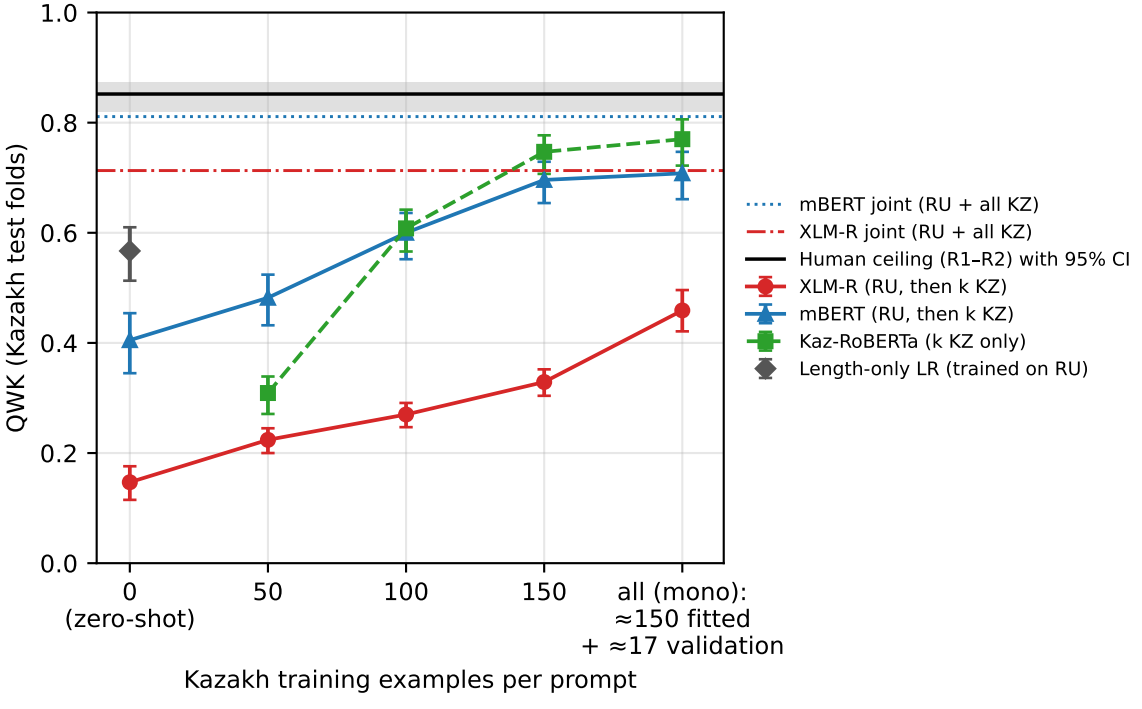
1.0
0.8
0.6
0.4
0.2
0.0
QWK (Kazakh test folds)
0
(zero-shot)
50
100
150
all (mono):
≈150 fitted
+ ≈17 validation
Kazakh training examples per prompt
mBERT joint (RU + all KZ)
XLM-R joint (RU + all KZ)
Human ceiling (R1–R2) with 95% CI
XLM-R (RU, then k KZ)
mBERT (RU, then k KZ)
Kaz-RoBERTa (k KZ only)
Length-only LR (trained on RU)